Pratibimb

THE HYPERTENSION TWIN

DIGITAL TWIN CHALLENGE 2026 | PHASE 1 | PROTOTYPE AND CODE

PRATIBIMB

प्रतिबिम्ब

Reflection, in Hindi and Sanskrit. A digital twin of one patient's body that fuses the health record with a budget watch and a home BP cuff, and warns the doctor up to 72 hours before a strain episode.

SUBMITTED BY

Team AMAZE

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IIM Lucknow, PGP in Sustainable Management

Live demo pratibimb-pi.vercel.app Code github.com/pndynkt/Team-AMAZE_IIM-Lucknow

HAPPIEST HEALTH

DIGITAL TWIN CHALLENGE 2026



Pratibimb
THE HYPERTENSION TWIN
DIGITAL TWIN CHALLENGE 2026

PRATIBIMB AT A GLANCE

A digital twin for Indian adults with high blood pressure, on the watch they already own

Read every night against the patient's own body. Pratibimb learns how one person responds to sleep and walking and alerts only on what those habits cannot explain: **70% of strain episodes caught, a median 48 hours early, at 1 false alert per 22 patient-days.**

SCAN FOR THE LIVE DEMO no sign-in

TEAM AMAZE
IIM LUCKNOW
PGPSM

01 WHAT WE FOUND

WHO WE BUILD FOR



Meera Iyer, 58 | Patient, Bengaluru — THE PATIENT
- Wants: to stay well and out of hospital
- Pain: feels fine until the day she does not
- Gets: one line in Hindi or English, her best habit



Dr Kavya Rao | Physician, 240 patients — THE DOCTOR
- Wants: to know whom to call today
- Pain: threshold alarms on half of all readings
- Gets: a risk-sorted list inside an alarm budget



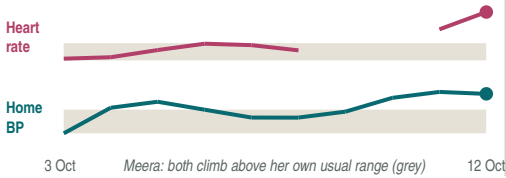
Anita Iyer, 34 | Daughter-in-law — THE FAMILY
- Wants: to know when to step in
- Pain: hears only once it is an emergency
- Gets: a consented SOS ladder, one tap to 112

THE CONDITION

24% of Indian men

21% of Indian women

have high blood pressure (NFHS-5). It is quiet: strain builds for days before a surge, a clinic or an ER visit.



Meera: both climb above her own usual range (grey)

WHAT THE EVIDENCE SAYS

- **WATCHES ARE EVERYWHERE:** 91.6% of smartwatches sold in India cost under US$50 (IDC). *So: build on the watch people own*
- **ALARMS DROWN DOCTORS:** fixed thresholds flagged 55.8% of readings in one study. *So: every alert sits inside a budget*
- **BODIES DIFFER:** in our cohort, an hour less sleep moves heart rate 0.3 to 2 bpm. *So: learn each person's own response*
- **A COLD LOOKS LIKE STRAIN:** heart rate rises in both; BP rarely does. *So: fuse the home BP cuff with the watch*

WHAT BLOCKS EARLY WARNING

- One rule for everyone
- Alarm fatigue
- Watches sync once a day
- Colds and late nights look alike
- No record context
- Privacy and trust

Each barrier maps to the answer that removes it (A1)

THE EVENT WE PREDICT

- **72 hours ahead** — a strain episode, scored every night
- **Against her own normal** — heart rate up, HRV down, sleep and steps down, home BP climbing
- **Amber, not red** — the model warns days ahead; SOS and 112 stay with people
- **Not a diagnosis** — decision support for a registered doctor

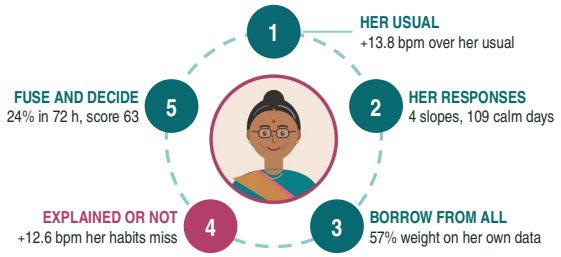
02 WHAT WE BUILT

THE WHITE SPACE: WHAT EACH APPROACH USES

	Health record	Watch	Home BP cuff	Own baseline	Own response	Calibrated	Alarm budget	Score
Threshold alarms		■	■					2/7
Overnight HR rule		■		■				2/7
Framingham score	■					▒		1.5/7
Same model, no twin	■	■	■	▒		■	■	5.5/7
PRATIBIMB	■	■	■	■	■	■	■	7/7

■ Uses it ▒ Partly □ Not used — Only Pratibimb fills all seven

HOW THE TWIN DECIDES, ON MEERA



Only the change her habits cannot explain becomes the warning

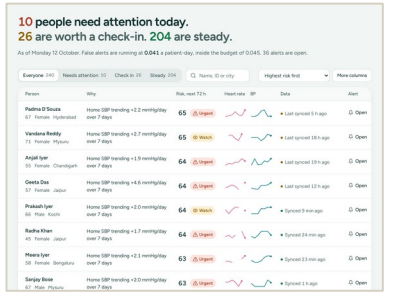
TWO STREAMS, ONE PERSON

HEALTH RECORD (STATIC)	WATCH AND CUFF (NIGHTLY)
Demographics	Resting heart rate, HRV
Past diagnoses (SNOMED CT)	Sleep stages: deep, REM
Lab results (LOINC)	Step count
Medicines and adherence	SpO2, breathing rate
Genetic marker, family history	Home BP cuff, mornings

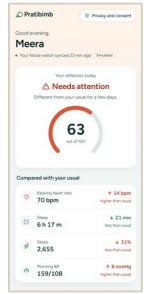
THE TWIN

240 synthetic patients x 120 days, 342 labelled episodes, seed 42

WHAT THE DOCTOR AND PATIENT SEE



Doctor: 240 people sorted by 72 h risk, reasons in words



Meera: one status, in her language

03 PROOF AND PATH

RESULTS ON PATIENTS THE MODEL NEVER SAW

	AUROC (0.5 is guessing)	Episodes caught at the budget
Pratibimb: all streams + twin	0.876	70%
All streams, no twin	0.851	64%
Watch + cuff + twin	0.870	72%
Watch only + twin	0.804	55%
Overnight HR rule	0.708	42%
Health record only	0.578	15%

Mean of 3 patient splits. Budget: 0.045 false alerts a patient-day.

THE TWIN LEARNS THE PERSON



Match with each hidden true slope (r)

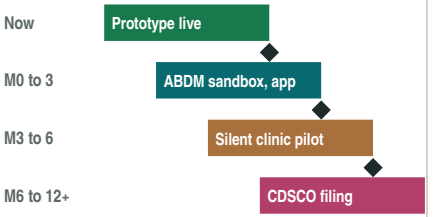
18 to 30% less error than one slope for everyone

ALERTS INSIDE THE BUDGET



1 false alert per 22 patient-days; median warning 48 hours before the event

GATED ROADMAP



Diamond = gate: each phase starts only when the last gate is met

BUILT TO BE TRUSTED

- **25 tests** — 3 check no future data leaks
- **MIT, public** — code, model card, live demo
- **4 consents** — one per stream, DPDP-ready
- **0 doses** — changed by the app, ever

THE ANSWER

Read every night against the patient's own body, alert only on what her habits cannot explain, and let people own the emergency. Decision rule for a pilot: proceed only if alerts stay inside the budget on real, consented data.

ASSUMPTIONS: all data is synthetic (240 patients x 120 days, seed 42); personas are illustrative; results are means of three patient splits on unseen patients.

SOURCE: NFHS-5 (J Health Popul Nutr, 2023); IDC India wearables, Q2 2026; arXiv 2603.09052; repository MODEL_CARD.md and ALGORITHM.md

A1 About 1 in 4 Indian men and 1 in 5 women have high blood pressure, and strain builds for days on watches nobody reads; the warning has to be personal and fit inside an alarm budget

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THE HYPERTENSION TWIN

Condition: hypertension with cardiovascular risk. Event: a strain episode starting within 72 hours. Personas are illustrative; Meera Iyer (P-0007) is a synthetic patient.

Condition in India | Use case

FOUR FACTS THAT SHAPE THE DESIGN

SEE S1 TO S4

24% | 21% — **of Indian men | women have high BP**
NFHS-5, 2019 to 21 (S1). Quiet and chronic: strain builds over days.
So: Pick hypertension; warn days ahead

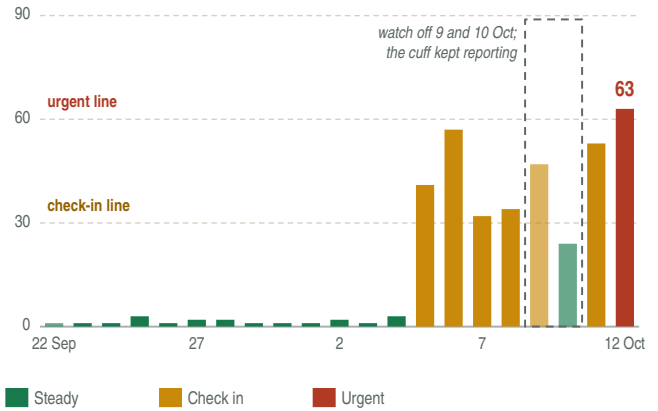
91.6% — **of smartwatches sold in India cost under US$50**
IDC, Q2 2026 (S2). They log heart rate, sleep and steps nightly.
So: Build on the watch people already own

55.8% — **of readings flagged by fixed thresholds**
One remote-monitoring study (S3). Doctors stop looking.
So: Every alert sits inside an alarm budget

0.3 to 2 — **bpm change in next-night heart rate per hour of sleep lost**
Person to person, in our cohort (S4). One rule cannot fit all.
So: Learn each person's own response

MEERA'S LAST 21 NIGHTS: RISK SCORE

SEE A5



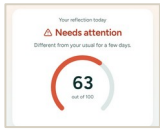
Score 0 to 100 from the trained model. The twin kept scoring while the watch was off, using the home BP cuff.

WHO WE DESIGN FOR

SEE A5



Meera Iyer, 58 | Patient, Bengaluru; hypertension, past heart attack
- **Wants:** to stay well and out of hospital
- **Pain:** feels fine until the day she does not
- **Gets:** one sentence in Hindi or English, her best habit, SOS



WHAT THEY SEE



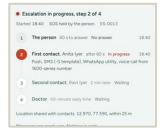
Dr Kavya Rao | Physician; 240 patients on remote watch
- **Wants:** to know whom to call today, and why
- **Pain:** threshold alarms on half of all readings
- **Gets:** a risk-sorted list inside an alarm budget



WHAT THEY SEE



Anita Iyer, 34 | Meera's daughter-in-law; first contact
- **Wants:** to know when to step in
- **Pain:** hears only once it is an emergency
- **Gets:** a consented SOS ladder with one tap to 112



WHAT THEY SEE

THE EVENT, DEFINED AGAINST HER OWN NORMAL

SEE A3

Signal	Rule (against her own baseline)	Meera on 12 Oct	Sign
Overnight resting heart rate	above her 28-night usual range	70.3 bpm vs 53.5 to 59.5	Up
HRV, 7-night mean	below her smallest worthwhile change	3.56 vs 3.67 to 3.80 lnRMSSD	Down
Home systolic BP	7-day trend climbing	159 mmHg, +2.1 mmHg a day	Up
Sleep and steps	falling below her usual week	6 h 17 m; 2,655 steps (31% under)	Down
Event day	BP surge, or an unplanned clinic or ER visit	episode under way	Event

Horizon: onset within 72 hours, scored every night from whatever the watch synced. **The model owns the amber tier;** SOS, family contacts and 112 own the red tier. We never claim heart-attack detection.

WHAT BLOCKS EARLY WARNING, AND THE ANSWER TO EACH

SEE A3 TO A6

One rule for everyone bodies differ 0.3 to 2 bpm	→	Her own baseline and response slopes — SEE A3
Alarms on half of readings doctors stop looking	→	Calibrated score inside an alarm budget — SEE A4
Colds and late nights look like strain on the wrist	→	Home BP cuff plus the twin-adjusted signal — SEE A3
Watches sync once a day no minute-level stream	→	Nightly scoring; freshness on every row — SEE A5
Emergencies need people no watch detects heart attacks	→	Consented SOS ladder, one tap to 112 — SEE A5
Privacy and trust health data, family contacts	→	Consent per stream; opt out of AI — SEE A6

SO WHAT **The data already arrives every night; what is missing is reading it against each person's own normal,** inside an alarm budget a doctor can live with.

ANNEXURE A1 | personas are illustrative composites; Meera, Kavya and Anita are synthetic; scores from the hosted demo

SOURCE: S1 NFHS-5 via J Health Popul Nutr 2023; S2 IDC India wearables Q2 2026; S3 arXiv 2603.09052; S4 cohort slopes, docs/MODEL_CARD.md

A2 Two synthetic streams fused into one person: a FHIR health record and a nightly watch-and-cuff stream for 240 patients, read by a twin engine behind one API

No real patient data anywhere: the generator is seeded (42) and hides each person's true responses so the twin can be checked; every field below is in the FHIR export.

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Data fusion | Synthetic data

THE STATIC RECORD (EHR)

MEERA, SYNTHETIC

Field	Standard	Meera Iyer
Demographics	FHIR Patient	58, female, Bengaluru; ABHA masked
Past diagnoses	SNOMED CT	Essential hypertension; IHD; MI (history)
Lab results	LOINC	HbA1c 5.8%; LDL 130; eGFR 83
Medicines	MedicationStatement	Amlodipine 5 mg; atorvastatin; aspirin
Adherence	dose log	missed amlodipine on 3 Oct
Genetic marker	polygenic risk tier	medium (simulated)
Family history	FamilyMemberHistory	father: heart attack at 61
Static risk	Framingham 2008	12.1% over 10 years (intermediate)

THE NIGHTLY STREAM (WATCH AND CUFF)

MEERA, 12 OCT

Signal	Source	Meera on 12 Oct
Resting heart rate	watch, 5-minute samples	70.3 bpm overnight
HRV (RMSSD)	watch, overnight	28.6 ms (ln 3.35)
Sleep and stages	watch: deep, REM, efficiency	6 h 17 m; 81% efficiency
Step count	watch	2,655 (usual about 4,000)
SpO2, breathing	watch	97%; 14.1 breaths a minute
Home BP	cuff, mornings	159/108 mmHg
Sync	Health Connect, HealthKit	hourly to daily; 23 min ago
CGM	not needed for BP	diabetes read from HbA1c

THE SYNTHETIC COHORT

SEE DATA_SOURCES

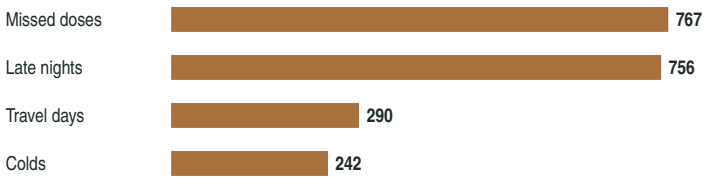
240 patients

120 days each

28,685 patient-days

342 strain episodes

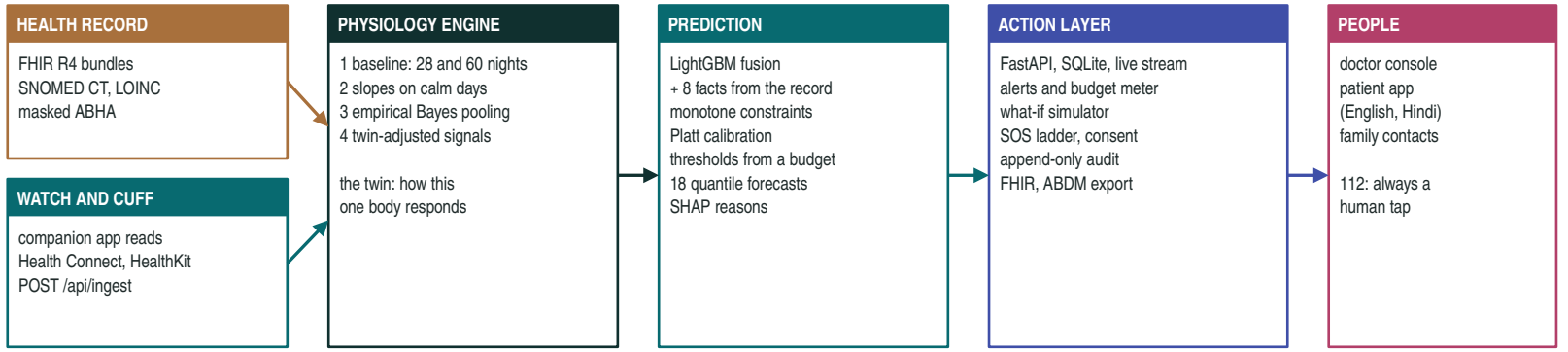
LOOKS LIKE TROUBLE, BUT IS NOT (PLANTED SO THE MODEL MUST TELL THEM APART)



Each person also carries hidden true responses to sleep and walking, so A4 can test whether the twin finds them.

ARCHITECTURE: TWO STREAMS IN, A WARNING AND A LADDER OUT

SEE ARCHITECTURE.PDF



TECHNICAL STACK

SEE README

Layer	Choice
Data	Python, NumPy, pandas; FHIR R4; seed 42
Twin engine	per-person regression, AR(1) errors, DerSimonian and Laird
Model	LightGBM, Platt scaling, SHAP, quantile forecasts
API	FastAPI, Pydantic v2, SQLite, server-sent events
Front end	React 18, TypeScript, Vite, Tailwind, Recharts
Standards	FHIR R4, ABDM WellnessRecord, SNOMED CT, LOINC
Quality	25 pytest tests, Docker, MIT licence

FUSION **The record does not just add columns: it tells the twin how to read the watch,** so a beta blocker or diabetes changes what a rise in heart rate means.

ANNEXURE A2 | all data synthetic; Meera's values are from the hosted demo on 12 Oct 2026

SOURCE: backend/twin/data (generator); docs/DATA_SOURCES.md; docs/DEVICE_SYNC.md; docs/architecture.pdf

A3 The twin compares tonight with Meera's own body: 13.8 bpm above her usual, only 1.3 explained by her habits, so 12.6 bpm becomes a calibrated 24% warning

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THE HYPERTENSION TWIN

Five steps, each with the trained model's real output for Meera Iyer on 12 October 2026. Every slope and baseline uses only earlier days (three automated leakage tests).

Predictive model | Personalisation

FIVE STEPS, WORKED ON MEERA

SEE ALGORITHM.MD

1 HER USUAL

Compares tonight with her own last 28 nights (heart rate, BP) and 60 nights (HRV), never with other people

+13.8 bpm

usual 56.5, tonight 70.3

features/build.py

2 HOW HER BODY RESPONDS

On calm days only, learns how tomorrow's heart rate, HRV and BP move with today's sleep and steps

4 slopes

from 109 calm days; errors widened for day-to-day drift

models/physiology.py

3 BORROW FROM THE GROUP

Pulls her slope toward the group by exactly how unsure it is (DerSimonian and Laird)

57%

weight on her own data; slope -1.41 vs group -1.03

models/physiology.py

4 EXPLAINED OR NOT

Subtracts what her own sleep and steps explain; recomputed daily from earlier days only

+12.6 bpm

unexplained; BP +16 mmHg her activity cannot explain

models/physiology.py

5 FUSE AND DECIDE

LightGBM on twin signals, BP trend and 8 record facts; calibrated; thresholds from the budget

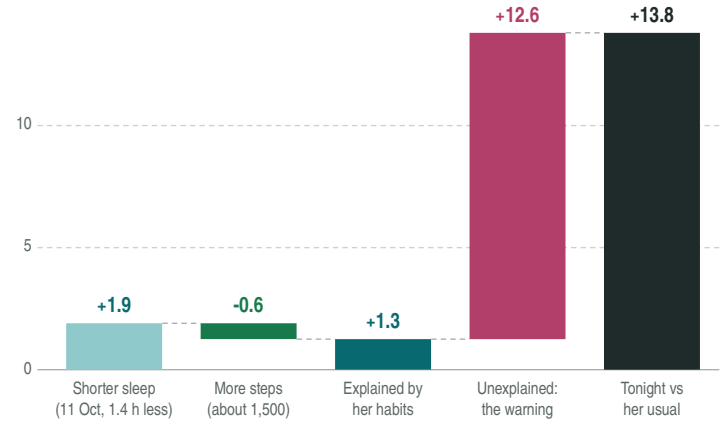
24%

in 72 h (likely 17 to 28%); usual 0.7%; score 63, urgent

models/fusion.py

STEP 4: WHERE TONIGHT'S +13.8 BPM COMES FROM

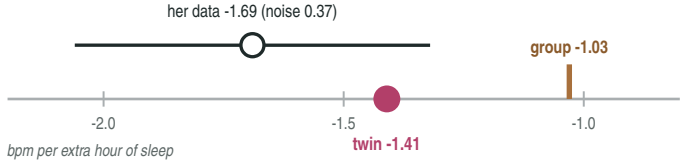
BPM



Explained = her slopes (fitted on earlier days only) x her habits: (-1.38) x (-1.38 h) + (-0.43) x (+1.50 thousand steps) = +1.3 bpm. **Home BP:** +16 mmHg above what her activity explains.

STEP 3: HOW MUCH TO TRUST HER OWN DATA

SLEEP TO HR



Weight on her own data = spread² / (spread² + noise²) = 0.43² / (0.43² + 0.37²) = 57%
Twin slope = 57% x (-1.69) + 43% x (-1.03) = -1.41



New users start at the group; the slope becomes theirs as calm days add up

STEP 5: FROM SIGNAL TO DECISION

FUSION.PY



24% chance of a strain episode in 72 hours (likely 17 to 28%); her usual 0.7%

What goes in	Guardrail
Twin-adjusted heart rate, HRV, BP	monotone: more sleep or walking never raises risk
7-day home BP trend; sleep debt; activity	Platt calibration: 20% means about 20%
8 record facts: Framingham, genetic tier, adherence, diabetes, prior MI, CKD, family history, beta blocker	urgent line = most sensitive cut within 0.045 false alerts a patient-day

Top reasons for Meera: home BP rising 2.1 mmHg a day; BP +16 mmHg unexplained; heart rate +13 bpm unexplained.

THE TWIN **A late night is explained by her own slope and stays quiet;** strain that her habits cannot explain becomes the warning.

ANNEXURE A3 | slopes are per extra hour of sleep or 1,000 steps; every value is computed by the trained model

SOURCE: docs/ALGORITHM.md; backend/twin/models/physiology.py and fusion.py; tests/test_leakage.py

A4 On patients the model never saw, the twin lifts AUROC from 0.851 to 0.876 and catches 70% of strain episodes a median 48 hours early, at 1 false alert per 22 patient-days

Pratibimb — THE HYPERTENSION TWIN

Patient-level 5-fold cross-validation, repeated on three patient splits (seeds 42, 43, 44); every number is the mean of the three. Synthetic cohort: real-world results will be lower.

Predictive model | Honest evaluation

ABLATION: WHAT EACH STREAM ADDS

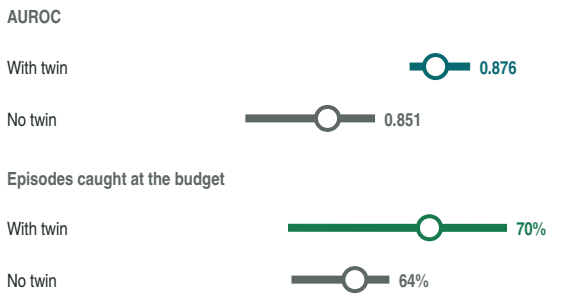
SEE MODEL_CARD

Model	Record	Watch	Cuff	Twin	AUROC	AUPRC	Caught at budget	48 h+
Pratibimb: all streams + twin	●	●	●	●	0.876	0.218	70%	50%
Watch + cuff + twin		●	●	●	0.870	0.214	72%	51%
All streams, no twin	●	●	●		0.851	0.189	64%	45%
Watch only + twin		●		●	0.804	0.136	55%	37%
Overnight HR rule (NightSignal-style)		●			0.708	0.091	42%	19%
Health record only	●				0.578	0.042	15%	15%

AUROC bar starts at 0.5 (guessing). The record raises ranking more than timing; its real job is reading the watch right.

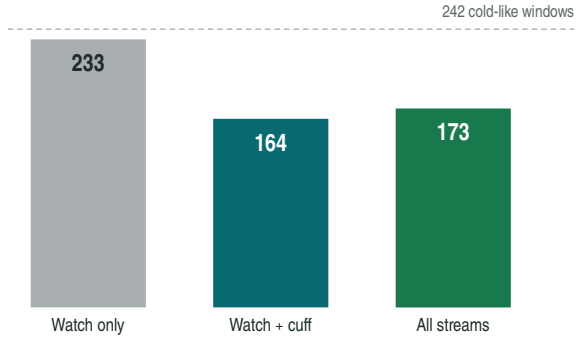
TWIN LIFT HOLDS ON EVERY SPLIT

MIN, MEAN, MAX



Better AUROC on 3 of 3 splits. Catch rate moves up to 19 points between splits of 240 people, so every headline is a three-split mean.

THE CUFF KEEPS COLDS FROM BECOMING ALARMS



Cold-like windows that raised an alarm, at 90% episode sensitivity. A cold raises heart rate; it rarely raises BP for a week.

DOES THE TWIN LEARN THE RIGHT PERSON?

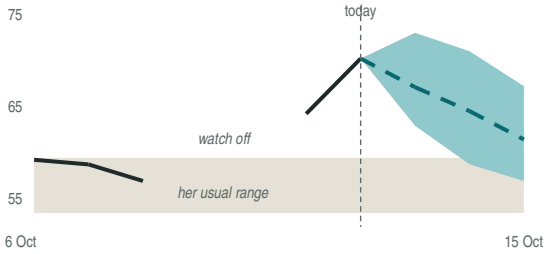
HIDDEN TRUTH KNOWN

Response the twin learns	Match with truth (r)	Twin error	One-slope error	Error cut	80% band holds
Sleep to next-night heart rate	0.71	0.28	0.40	30%	72%
Walking to next-morning BP	0.67	0.30	0.42	28%	74%
Walking to next-night heart rate	0.66	0.09	0.12	25%	76%
Sleep to next-night HRV	0.59	1.32	1.60	18%	75%

The generator plants each person's true slopes, so recovery can be checked directly. Error is mean absolute error in each slope's own unit; one-slope uses the population average for everyone.

NEXT 3 DAYS: MEERA'S HEART RATE (BPM)

P10 TO P90



Tomorrow 67 bpm (likely 63 to 73). **Day-1 error:** 2.5 bpm heart rate, 6.2 mmHg BP; 73% of readings land inside the P10 to P90 band.

HONEST LIMITS

- **Synthetic labels:** real-world results will be lower
- **Published anchors:** NightSignal 80% sensitivity (infections); EDI AUROC 0.76
- **Nightly, not live:** budget watches sync hourly to daily
- **Not heart-attack detection:** acute events go to SOS and 112
- **Record adds ranking:** not much timing

PROOF **The lift comes from the twin on every split, and the cuff keeps colds from becoming alarms;** these are synthetic results, so the next test is a real clinic.

ANNEXURE A4 | budget = 0.045 false alerts a patient-day; 342 episodes in 28,685 patient-days; three patient splits

SOURCE: docs/MODEL_CARD.md; backend/artifacts/model_card.json; tests/test_leakage.py; NightSignal and EDI as cited in the model card

A5 The doctor sees who needs attention and why, tries changes on the patient's own body and acts; the patient gets one plain sentence, her best habit and a safe SOS ladder

Live at pratibimb-pi.vercel.app with no sign-in; the online demo replays real output from the trained model on the synthetic cohort.

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THE HYPERTENSION TWIN

Doctor dashboard | Patient app

TODAY: WHO NEEDS ATTENTION

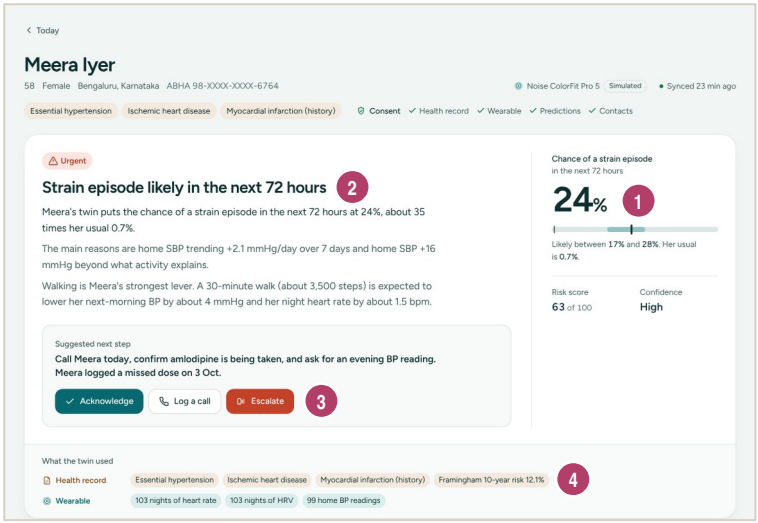
DOCTOR CONSOLE



1. **10 urgent, 26 check-in, 204 steady** out of 240 people on 12 Oct
2. **Sorted by 72 h risk** with the reason in plain words
3. **Trend lines** heart rate and home BP, last week
4. **Data freshness** budget watches sync hourly to daily

TWIN PAGE: WHY, AND WHAT TO DO

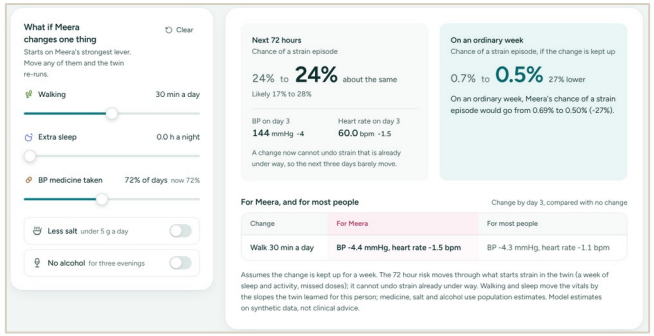
MEERA'S PAGE



1. **24% in 72 hours** likely 17 to 28%; her usual 0.7%
2. **Three sentences** the chance, the reasons, her best lever
3. **Next step** call Meera, confirm amlodipine, evening BP
4. **What the twin used** record and watch, side by side

WHAT IF: ON HER OWN SLOPES

SIMULATE TAB



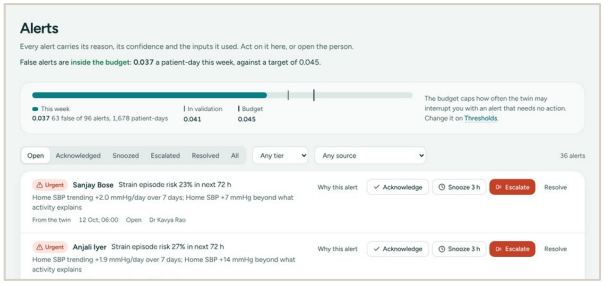
-4.4 mmHg
BP with a daily 30 min walk, on her slope (most people -4.3)

27% lower
ordinary-week risk: 0.7% to 0.5%

holds at 24%
next 72 h: the strain is already under way

ALERTS INSIDE A BUDGET

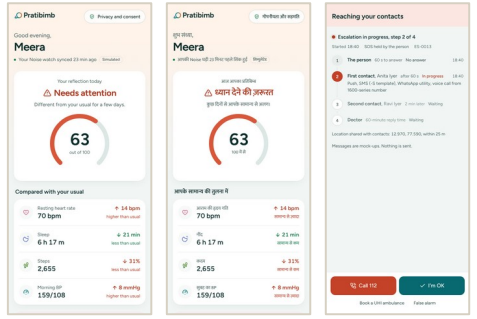
ALERTS PAGE



0.037 false alerts a patient-day this week vs a **0.045** budget. Any threshold change is replayed on 14 days first; every action goes to an append-only audit log.

MEERA'S PHONE AND THE SOS LADDER

PATIENT VIEW



Status | Hindi, read-aloud | SOS ladder

1. Meera, 60 s to answer
2. first contact
3. second contact
4. her doctor

One tap to 112 at every step; the server never dials it.

IN PRACTICE **The doctor spends time on the 10 who need it, not on all 240;** the patient hears one sentence in her language and knows what to do.

ANNEXURE A5 | screens from the live demo on synthetic data; messages to contacts are mock-ups and say so on screen

SOURCE: pratibimb-pi.vercel.app; docs/API.md; docs/COMPLIANCE.md

A6 Built for Indian practice and rules, honest about its limits, with a gated path from this prototype to a silent clinic pilot and a CDSCO filing

What runs today versus what a production version adds. Phases and gates are proposed: each phase starts only when the gate before it is met.

Pratibimb
THE HYPERTENSION TWIN

Indian impact | Roadmap

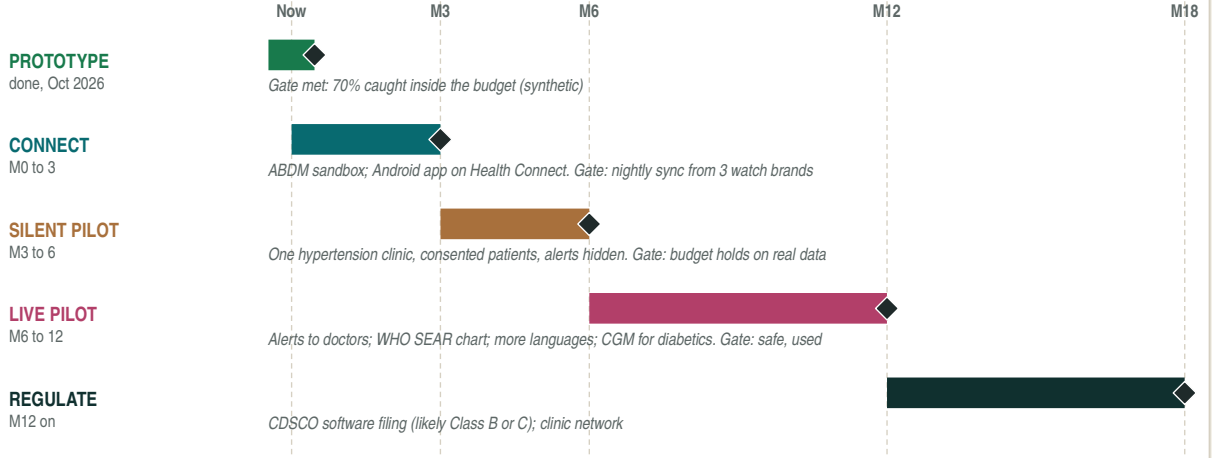
COMPLIANCE, MAPPED TO WHAT RUNS TODAY

SEE COMPLIANCE.MD

Rule	What Pratibimb does today	Production adds
DPDP Act 2023	consent per stream; opt out of AI; withdraw and delete; 18+ only	privacy notice, grievance officer, India hosting
Telemedicine Guidelines 2020	never changes a dose; the doctor decides	clinician sign-off workflow
CDSCO software guidance, 2026	model card, synthetic-data disclosure, audit log	likely Class B or C; QMS; clinical validation
ABDM	FHIR R4 and WellnessRecord export; ABHA masked	sandbox milestones M1 and M3; WASA audit
TRAI commercial messages	SMS, WhatsApp and voice templates (mocked)	DLT registration of headers and templates
ICMR AI ethics 2023	patients can refuse AI; reasons always shown	bias benchmarking before deployment

FIVE PHASES, EACH WITH A GATE

PROPOSED



BUILT FOR INDIA

- **Budget watches** via Android Health Connect; Apple Watch via HealthKit
- **Home BP cuff** the cheapest sensor that separates strain from a cold
- **English and Hindi** with read-aloud on the patient view
- **One tap to 112** and an ambulance booking link at every SOS step
- **ABDM ready** WellnessRecord export; ABHA masked everywhere

HONEST LIMITS

- **Synthetic labels:** every result is on generated data; real numbers will be lower
- **Nightly, not minute by minute:** budget watches sync hourly to daily
- **No heart-attack detection:** acute events go to SOS, family and 112
- **Record adds ranking, not timing:** its job is reading the watch correctly
- **Framingham under-estimates South Asians:** the WHO 2019 SEAR chart is next

TEAM AMAZE, IIM LUCKNOW

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LIVE DEMO CODE

Live demo pratibimb-pi.vercel.app
Code github.com/pndynkt/Team-AMAZE_IIM-Lucknow
Read first README, docs/ALGORITHM.md, docs/MODEL_CARD.md

Research prototype on synthetic data. Not a medical device. In an emergency, call 112.

NEXT **A silent pilot with one hypertension clinic on consented real data;** the alarm budget and the twin's recovery of each person are the gates.

ANNEXURE A6 | phases, months and gates are the team's proposal; regulatory class is a judgement pending CDSCO review

SOURCE: docs/COMPLIANCE.md; DPDP Act 2023; CDSCO Medical Device Software guidance, 30 Jul 2026; NRCeS ABDM guide

Pratibimb

THE HYPERTENSION TWIN

THANK YOU

Read the body you have, not the average one.

HAPPIEST HEALTH

DIGITAL TWIN CHALLENGE 2026



ANNEXURES: THE PROOF BEHIND THE ONE-PAGER

A1 **The problem and the people**
four facts, Meera's 21 nights, personas, the event, barriers

A2 **Data and architecture**
the record, the nightly stream, the cohort, the stack

A3 **How the twin decides**
five steps on Meera, the bridge, borrowing strength, the decision

A4 **Does it work**
ablation, split ranges, colds vs the cuff, recovery, forecasts, limits

A5 **The product**
doctor console, twin page, what-if, alert budget, phone and SOS

A6 **India, safety and the road ahead**
compliance, gated phases, India fit, limits, team and links

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